**SMARTROUTEAI**

**EXPERIMENT NO. 3: GREEDY BEST-FIRST SEARCH AND A* SEARCH**

**Course:** Artificial Intelligence Programming
**Group:** 9
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**1. Objectives**

- Implement Greedy Best-First Search and A* Search.
- Demonstrate intelligent route finding.
- Compare search algorithms using an interactive dashboard.

**2. Technologies Used**

- Python 3.13 – implementation.
- Streamlit – interactive dashboard.
- OSMnx and OpenStreetMap – real-world road networks.
- Folium – interactive route maps.
- Python priority queues and graph data structures – search implementation.

**3. Algorithms Implemented**

- **Greedy Best-First Search:** Uses heuristic (h(n)) to select the next node.
- *A Search:** Uses (f(n)=g(n)+h(n)) to select the next node.
- **Dijkstra's Algorithm:** Additional benchmark for shortest-path verification.

**4. Implementation**

- Created a Romania road-network graph.

- Implemented Greedy Best-First Search and A*.
- Developed an interactive Streamlit dashboard.
- Added route visualisation and performance comparisons.
- Integrated OpenStreetMap for real-world routing.
- Added interactive maps and Dijkstra verification.

**5. Testing and Results**

**A. Romania Road Network – Arad to Bucharest**

- Greedy: 450 km.
- A*: 418 km.

**B. Real-World Routing – Nairobi**

- Test route: Westlands to Jomo Kenyatta International Airport.
- Greedy: approximately 34.01 km.
- A*: approximately 27.05 km.
- Dijkstra: approximately 27.05 km.
- A* matched Dijkstra's shortest distance in this test.

**6. Limitations**

- Real-world routing has been reliably tested within Nairobi.
- Intercity routing, such as Vihiga to Nairobi, is not yet reliably supported.
- The real-world module depends on external OpenStreetMap services.

**7. Conclusion**

- Successfully implemented both required search algorithms.
- Demonstrated their operation through an interactive dashboard.
- Extended the project to real-world road routing.
- Verified an A* result against Dijkstra's algorithm.

**8. Future Improvements**

- Improve intercity and worldwide routing.

- Add location autocomplete.
- Support additional transport modes.
- Improve deployment and performance.

**9. References**


- https://www.geeksforgeeks.org/best-first-search-informed-search/
- https://www.geeksforgeeks.org/a-search-algorithm/
- https://www.openstreetmap.org/
- https://osmnx.readthedocs.io/